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Identifying syntactic biomarkers of cognitive impairment in Mandarin-speaking older adults by applying machine learning approaches across multiple speech tasks

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ABSTRACT

Background: Dementia can impair cognitive functions in older adults and further affect language abilities, such as sentence construction. The use of linguistic biomarkers for detecting cognitive decline and early stages of dementia has demonstrated great potential due to its low-cost and non-invasive nature, especially towards a large aging population.

Aims: In this study, we focused on the syntactic complexity and aimed to identify highly discriminative syntactic features and those set in differentiating Mandarin-speaking older adults with and without cognitive impairment using machine learning models.

Methods & procedures: We established a new cohort consisting of 52 Chinese older participants, which has not been reported in any previous studies. After applying exclusion criteria, 49 were selected for this study, including 24 labeled as cognitively normal (CN) and 25 as cognitively impaired (CI) based on their scores in the MoCA-B test (Chinese version). Each of the participants completed three connected speech tasks: a picture description task (the Picnic Scene), a story narrative task (*Sanmao's Wanderings*), and a story recall task (Cowherd and Weaver Girl). The speech recordings were automatically transcribed into text files and manually checked. We then performed treebank annotation, which was also manually corrected. Subsequently, 17 linguistic unit-based and 10 syntactic structure-based features were extracted from the texts using Python codes. To identify key syntactic features and feature sets, we first examined the discriminative power of these features per se for distinguishing between the CI and CN groups. Then we employed three machine learning algorithms to identify feature combinations performing the best in terms of high accuracy and AUC in cross-validation and minimal number of basic features.

Outcomes & results: No single feature or feature panels showed high discriminability between groups across all three tasks. However, certain macro-level features with high measurement robustness, such as mean depth of nodes, mean

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dependency distance, and the ratio of head-final dependencies in specific tasks are noteworthy. Moreover, through machine learning algorithms, we identified several well-performing feature combinations in terms of the metrics of cross-validation mean accuracy and area under curve (AUC). Notably, a feature panel with only two features (mean depth of nodes, and proportions of prepositional dependencies) achieved the best result across classifiers in the story narrative task (ACC = 0.840, AUC = 0.867).

Conclusions: This study demonstrated the potential of using purely treebank-derived features within a single linguistic domain (syntax) for cost-effective cognitive impairment screening, indicating the value of syntactic biomarkers for cognitive impairment in Mandarin-Speaking Older Adults.

Introduction

Aging-related cognitive decline presents a significant global health issue that has garnered substantial public attention and research focus. Typical approaches to the detection of cognitive impairment (CI) and Alzheimer's disease (AD) include utilizing sensitive neuroimaging (Cole & Franke, 2017; Soch et al., 2021) and fluid biomarkers such as levels of amyloid β (A β) and tau proteins (Blennow et al., 2010; Jack et al., 2013; Olsson et al., 2016). Despite the considerable volume of research on these aging-related biomarkers and the development of diagnostic frameworks (Higgins-Chen et al., 2021; Y.-J. Jia et al., 2023), their clinical utility for large scale cognitive screening in aging population is often limited due to the invasive and resource-intensive nature of the medical procedures. This limitation hampers their wide-scale implementation in community-based screening or monitoring programs. Given the drawbacks of the aforementioned biomarker detection in medical settings, using connected speech has risen as an emerging approach, owing to their non-invasiveness, cost-effectiveness, and accessibility (Ahmed et al., 2013; Ash & Grossman, 2015; Mueller et al., 2018). Several authoritative diagnostic criteria for dementia and cognitive impairment (DSM-5, American Psychiatric Association, 2013; IWG-2; Dubois et al., 2014; Chinese Diagnosis and Treatment of Dementia and Cognitive Impairment Guide; J. Jia, 2016) all regard linguistic features as important clinical biomarkers. Evidence suggests that alternations in spoken language happen at an early stage of cognitive decline, in terms of acoustic and linguistic features (Guinn et al., 2014; Hason & Krishnan, 2022; He et al., 2023). Recently, the development of automatic speech processing and natural language processing tools has greatly facilitated the process of automatic feature extraction from speech data and transcribed texts (Martínez-Nicolás et al., 2021). In this paper, we aim to further advance research on distinguishing cognitive impairment from healthy controls based on connected speech.

Linguistic biomarkers of cognitive impairment

Thus far, several linguistic domains have been explored extensively, including phonetic fluency, lexical richness, and semantic content. Commonly used acoustic features include pitches, jitters, shimmers, mel-frequency cepstral coefficients (Al-Hameed et al., 2017;

Yamada et al., 2023), while lexical and semantic measures contain the number of word types, part-of-speech tags, propositional idea density (Fraser et al., 2016; Yamada et al., 2023).

It is noteworthy that another key feature of cognitive impairment lies in the decline in syntactic complexity, reflecting the language users' competence to compose complex sentences properly and to use sophisticated constructions (Bulté & Housen, 2012; Ortega, 2003). It is deemed to be correlated primarily with working memory, which plays a role in the process of sentence production (Cheung & Kemper, 1992; Fleming & Harris, 2008; Roark et al., 2007). Pioneering investigations, such as the classical Nun Study by Snowden et al. (1996) and a longitudinal study by Kemper et al. (2001), have revealed the correlation between syntactic complexity and aging-related cognitive decline.

In recent years, rapid advancements in the techniques of natural language processing, such as parsing, have facilitated automatic analysis of syntactic complexity. Consequently, researchers extensively explored the potential of syntactic features as a domain of biomarkers for dementia detection. Related studies fall into two categories. The first category focuses on micro-level, specific constructions, and the key technical challenge lie in the accurate identification of specific phenomena. For instance, de Lira et al. (2011) and Orimaye et al. (2014) paid attention to three specific types of constructions (coordinated, subordinated, and reduced sentences), while Orimaye et al. (2017) used a huge number of up to 16,926 lexical and syntactic features, with the majority being n-gram features. Sung et al. (2020) found that passive structures are the best predictor of mild cognitive impairment in sentence comprehension. The second category utilized macro-level, holistic measures of syntax, which primarily rely on the results of syntactic parsing. For examples, Roark et al. (2007) and Pakhomov et al. (2011) took an automatic assessment approach to the exploration of syntactic-tree based measures such as Frazier score, Yngve score, and mean dependency distance (MDD). Lundholm Fors et al. (2018) further explored several variants of MDD. J. Liu et al. (2021), on the other hand, applied the complex network parameters of syntactic networks as novel measures.

Ye and Huang (to appear) conducted a dedicated systematic review of syntactic indicators of normal aging and cognitive impairment in speech production of older adults (primarily English-speaking) over thirty studies focusing on the trend of specific research question, and summarized a list of important syntactic measures and syntactic constructions, which were shown to be significantly correlated with the above-mentioned two factors across various studies. For cognitive decline, the key measures include mean length of utterance, mean clauses per utterance, mean dependency distance, and developmental level, while the key syntactic constructions comprise verbs/predicates, reduced sentences, left-branching structures and coordination. They ended their paper with the conclusion that the use of syntactic measures in cognitive decline detection thus far is still limited, and that more candidates should be put into tests for their sensitivity.

However, there are several deficiencies in this line. Some studies only reported *p*-values but not effect sizes or AUC values (the area under the Receiver Operating Characteristic curve),¹ a common metric to quantify the discriminant power of two groups on a certain dimension, which weakens the efficacy of these studies and renders them less useful for reference and comparison in subsequent research (e.g., Roark et al., 2007). Another issue is that different studies sometimes reported contradictory tendencies in the change of measures or different significance levels regarding certain syntactic features, such as

MDD, which needs further verification. Thirdly, only a few syntactic complexity measures have been explored sporadically in the research taking cognitive decline as a variable. As a reference, among those studies focusing on the effect of natural aging, the early and classical study of Cheung and Kemper (1992) compared 11 syntactic complexity measures of speech productions older adults, and likewise, the recent study of Agmon et al. (2024) supplemented several up-to-date syntactic measures and utilized 8 measures in all. However, we are aware of no dedicated, comprehensive study focusing solely on the syntactic features between cognitively impaired and cognitively healthy older adults who speak Mandarin.

In this study, we intend to consider more candidates of syntactic measures to assess their ability to distinguish between cognitively healthy individuals and those with cognitive decline.

Machine learning approaches to cognitive decline detection

In addition to using individual features as sensitive biomarkers of cognitively impairment, the machine learning (ML) techniques offer a potent tool for disease detection, enabling more accurate classification based on multiple variables. Given that ML models are in nature mathematical combinations of input features, they can facilitate the search for composite biomarkers of diseases. For instance, Meyer et al. (2024) proposed a composite measure called the Amyloid Probability Score 2 (APS2) as a combination of the plasma % p-tau217 and the A β 42/40 ratio, using the linear model of logistic regression. There have been a large number of studies utilizing ML algorithms for the detection of dementia and Alzheimer's disease based on neuroimaging (Goenka & Tiwari, 2021), fluid (Blanco et al., 2023), and multimodal biomarkers (Javeed et al., 2023). As for speech- or text-based detection of early cognitive decline, such method has also been shown to be effective (de la Fuente Garcia et al., 2020; Nasreen et al., 2021; Huang et al., 2024; Shah et al., 2021). Currently, there are two main approaches.

The first approach utilized handcrafted features with traditional machine learning classifiers. In terms of modality, some studies rely on the speech modality (Gosztolya et al., 2021; R. Wang et al., 2023), others on transcribed text (Chen et al., 2024; Orimaye et al., 2017), while some combine both (He et al., 2023; Yamada et al., 2023). Certain studies have compared features across these two modalities (Huang et al., 2024), but thus far, there is no definitive conclusion regarding which modality produces better models. The number of features employed in these studies range from a few (Chen et al., 2024, p. 8 features) to tens (Asgari et al., 2017, p. 68 features), hundreds (Fraser et al., 2016, p. 370 features), thousands (He et al., 2023, p. 7354 features), or even tens of thousands of features (Orimaye et al., 2017, p. 16926 features), which are often first selected and then combined using machine learning models. As for classifiers, most commonly used ones include support vector machines (SVM), logistic regression (LR), random forests, and naïve Bayes. These models can be categorized into two types. One type consists of interpretable linear models (e.g., SVM, LR), which, however, often exhibit weaker classification performance when used alone. The other type includes black-box models, such as random forests and XGBoost, which achieve better performance but struggle to represent the final feature dimension used for classification as a simple function of basic features.

The second is the deep learning approach, which utilizes neural network models. Huang and Ye (2024) conducted a systematic review of studies that employed deep learning methods for cognitive status screening based on speech or language. They categorized such studies into five types according to the role that neural networks play in the research process, namely as a data augmentation tool, as a preprocessing tool, as a pre-trained model for extracting initial embedding features, as a feature extractor for high-level feature extraction, and as the final classification head. Among these, the last type can be considered part of traditional machine learning, which simply uses a multilayer perceptron as the classifier. Overall, they also concluded that due to the limited availability of data from potential older adults with dementia, neural networks employed for end-to-end training in this field remain relatively scarce. Therefore, despite the dominance of deep learning, traditional machine learning methods with handcrafted features of linguistic significance remain important.

In this study, since our goal is to identify linguistic biomarkers of cognitive impairment, we prioritize the interpretability of feature (combinations). Therefore, we focus on the application of interpretable linear machine learning models to assess whether satisfactory results can be achieved.

Cognitive decline detection through connected speech in the Mandarin-speaking population

China has over 15 million people with dementia, accounting for approximately 26% of the global dementia population. This number is projected to continue rising and is expected to reach 45 million by around 2050, representing 29% of the global total (2022). Developing low-cost, language-based cognitive screening for native Mandarin speakers is crucial for early detection and intervention. However, a review of the cognitive decline detection through connected speech showed that the vast majority of the studies were conducted on the cohorts of English-speaking older adults (Huang & Ye, 2024), while only a small number focused on the native speakers of Mandarin.

Among these, in terms of the dataset employed, some researchers put emphasis on the development of new algorithms, and utilized existent Chinese Alzheimer's disease detection challenge datasets, including NCMMSC2021 (Ke et al., 2024; Yang et al., 2022; Ying et al., 2023), iFLY (N. Liu et al., 2022), etc. Of the abovementioned studies, Ke et al. (2024) yielded the highest accuracy of 87.1% using deep learning with an elaborate feature selection algorithm. Other research teams experimented on their self-constructed datasets. Data were collected in Chinese Taiwan (Chien et al., 2019; Tsai et al., 2021), Changsha (R. Wang et al., 2023), Shanghai (Huang et al., 2024), Yunnan (J. Liu et al., 2021), Shenzhen (T. Wang et al., 2019), and Beijing (T. Wang et al., 2021).

For the approach taken, those research teams from linguistics mainly applied traditional machine learning models to manually extracted phonetic and linguistic features (e.g., Huang et al., 2024; R. Wang et al., 2023), while those from computer science and artificial intelligence department have primarily employed deep learning methods (e.g., Ke et al., 2024; Ying et al., 2023).

As for the model performance of these studies, T. Wang et al. (2021) have achieved the highest scores, with ACC = 0.96 and F1 = 0.96. They employed three speech tasks (picture description, semantic fluency, sentence repetition), and the final model with late fusion

based on soft voting performed better than each of the classifiers trained on single tasks. Focusing on the performance of single classifiers, sentence repetition yields best results, approaching 0.9, while the other two (picture description, semantic fluency) are still below 0.9, just like other studies. However, their study presents two critical problems. First, their approach relies on a large set of 90 features from various domains, each requiring different computational methods, making replication highly challenging and demanding extremely stringent conditions. In addition, their reported accuracy of 96%, achieved solely through linguistic biomarkers, is significantly higher than that of all other known studies in the field and almost approaches the performance of AD core biomarkers, which seems counterintuitive. Given that no follow-up replication attempts have been made by their research team or others, these results remain questionable and difficult to be translated into practical applications or cognitive screening. Excluding T. Wang et al. (2021), most of the best metrics (accuracy, F-score or AUC) in the remaining studies fall between 0.8 and 0.9. Studies employing deep learning have not demonstrated significantly superior performance, which may be attributed to the limited data size.

Overall, the research on the Mandarin-speaking population is still insufficient. It remains unclear which measures are crucial for connected speech-based cognitive decline detection in Chinese older adults, and whether the critical markers found effective in other languages are also applicable to Mandarin. This is particularly the case for syntactic complexity measures, which remain largely understudied.

The present study

To address the limitations of previous studies, the current study aims to identify key features or feature combinations effective for detecting cognitive decline with special focus on syntactic complexity, a key aspect of connected speech, thereby contributing to the growing body of research on the linguistic biomarkers of early dementia, and pioneering this paradigm in a Mandarin context.

Specifically, the present study aims to address the following research questions:

- (1) Which models and feature combinations demonstrate high discriminability between cognitively impaired and cognitively healthy populations? How do these syntactic complexity-based models perform compared with those reported in previous literature?
- (2) Among these models, which syntactic feature(s) play significant roles?
- (3) Are there any features or feature combinations that exhibit high discriminability across different tasks and classifiers?
- (4) Taken together, how do we determine the optimal feature combination?

To achieve this, we utilized a newly established cohort of Mandarin-speaking older adults (cognitively normal, $n = 24$; cognitively impaired, $n = 25$), and extracted 27 syntactic complexity measures from their transcribed speech elicited by three speech tasks (picture description, story narrative, and story recall). We then applied feature selection algorithms to select highly discriminative features, employed multiple linear machine learning models to differentiate between the cognitively impaired population and healthy controls, and compared them across the three tasks.

Our study possesses the following novelties. First, the cohort and corpus data used in this study were newly collected and have not been reported in any prior research. Second, all our features were derived solely from the syntactic domain and extracted entirely based on annotated treebanks in a rule-based fashion, making replication easier compared to studies that utilize multi-domain features. Finally, we employed interpretable linear models, which are more intuitive as the final composite features can be represented as a weighted linear combination of the input features.

Materials and methods

Participants

The dataset used in this study was selected from the Multimodal Corpus of Gerontic Discourse (MCGD), one of the most prominent Mandarin corpora for research on aging and language (Zhou, 2024). No power analysis was conducted to pre-determine the sample size, but our sample size is comparable to those in similar publications (Clarke et al., 2021). A total number of 52 participants were recruited from 10 local communities in Shanghai. Participants were recruited through multiple community-based approaches, including the distribution of recruitment posters, the organization of public service lectures, and collaboration with local comprehensive service centers for older adults and subdistrict associations for the advancement of aging services. During recruitment, the research team provided potential participants with detailed information regarding the study's purpose, procedures, and potential risks and benefits, ensuring that participation was entirely voluntary and based on informed consent. Older community residents or their family members who expressed interest in the study were encouraged to contact the research team for further information or enrollment. The study was conducted under the approval of the Research Ethics Committee of Tongji University (ethical review number: tjslfrec202306). All participants provided written informed consent to participate in the study. All participants retained basic functional abilities for daily living, and did not have Parkinson's disease or psychiatric disorders. Other neurological conditions were excluded based on a structured medical history interview, during which participants were asked to report any diagnoses, treatments, or symptoms related to neurological disorders. Only individuals without such medical histories were enrolled in the study. All of them were fluent native speakers of both Mandarin and Shanghainese. Except for two participants who had received no education and one participant who failed to recall the exact years of education (YoE), the YoE of other participants ranged from 6 to 15. The speech data of three of the recruited participants were excluded from this study for reasons described in the following section, resulting in a sum of 49 participants.

To assess the cognitive ability of the participants, the participants were first instructed to complete a Montreal Cognitive Assessment – Basic test (MoCA-B, Chinese version).² Five graduate students majoring in applied linguistics were recruited and trained by the neurology department of a Grade-A tertiary hospital in Shanghai to give standard instructions to the participants. Our classification criteria are based on Huang et al. (2018). The cognitive levels of the participants were derived based on their scores for different levels of highest education,

Table 1. Demographic information and cognitive measurements of the participants.

Characteristics	CN (N = 24)	CI (N = 25)	<i>t</i> or χ^2 statistic	<i>p</i> value
Age (years)	72.04 (6.61)	72.76 (7.74)	0.3416	0.7342
Sex (female)	17 (73.83%)	20 (80.00%)	0.1711	0.6791
Education (years)	10.30 (3.89)	10.21 (2.20)	−0.0911	0.9278
MoCA-B scores	25.21 (2.66)	15.64 (5.70)	−7.3260	<0.0001***

The age, education, MoCA-B scores are shown as means (standard deviations), and the *p*-value assessed with the *t*-tests. The participant who failed to recall the years of education is ruled out. The sex data are shown as the number and percentage of female participants, with the *p*-values assessed with a chi-square test. The parentheses following the “CN” and “CI” head indicate the number of participants in each of the two groups. *** indicates high significance.

measured by the YoE. The MoCA-B scores that distinguish a normal control person from a participant with cognitive impairment (including mild, moderate, and severe AD) are as follows: 20 for less than 7 YoE; 23 for 7 to 12 YoE; 25 for over 12 YoE. [Table A1](#) summarizes the full criterion for assessing the cognitive ability of a participant in a MoCA-B test.

The demographic information and MoCA-B scores of the participants are listed in [Table 1](#). Among the 52 participants we recruited, 25 were identified as cognitively normal (CN), 19 as mild cognitive impairment (MCI), 3 as mild dementia, and 5 as moderate or severe dementia. The diagnosis was made by neurologists based on a clinical assessment, with supplementary information from MoCA-B scores to support the diagnosis. The participants labeled as MCI and different levels of dementia were grouped as cognitive impairment (CI) in the following ML classification. After collecting the response of the participants in each of the speech tasks, we excluded three participants who were not familiar with the materials employed in the story-telling task and we finally obtained 24 participants labeled as CN and 25 participants labeled as CI. We acknowledge that, from a machine learning perspective, the sample size used in this study is relatively small. However, given the challenges in recruiting individuals with cognitive impairment, we followed the approach of previous studies with similarly small sample sizes (Gosztolya et al., 2021; Liang et al., 2022; Pistono et al., 2021).

Speech data collection and data processing

[Figure 1](#) shows the whole data processing and analytical process. First, the participants were asked to complete three speech tasks. The recordings were then automatically transcribed into text files using the Feishu toolkit, an online open-source platform for audio transcription with high accuracy for Mandarin (ByteDance, 2023). We segmented each transcription into three files according to each speech task. Spoken discourse phenomena, such as fillers, self-repairs, repetitions, and incomplete sentences, were also transcribed but systematically removed before parsing. These transcription segments were then processed using Stanza (version 2.0) package, a Python-based Natural Language Processing tool, for sentence segmentation, part-of-speech tagging, dependency parsing, etc (Qi et al., 2020). Manual corrections were conducted at two stages: after the initial automatic transcription and again during treebank annotation. The corrections were performed by two linguistics PhD students, and any discrepancies were resolved through discussion to ensure consistency. Finally, automatic linguistic unit identification and feature extractions were done by self-compiled python codes.³

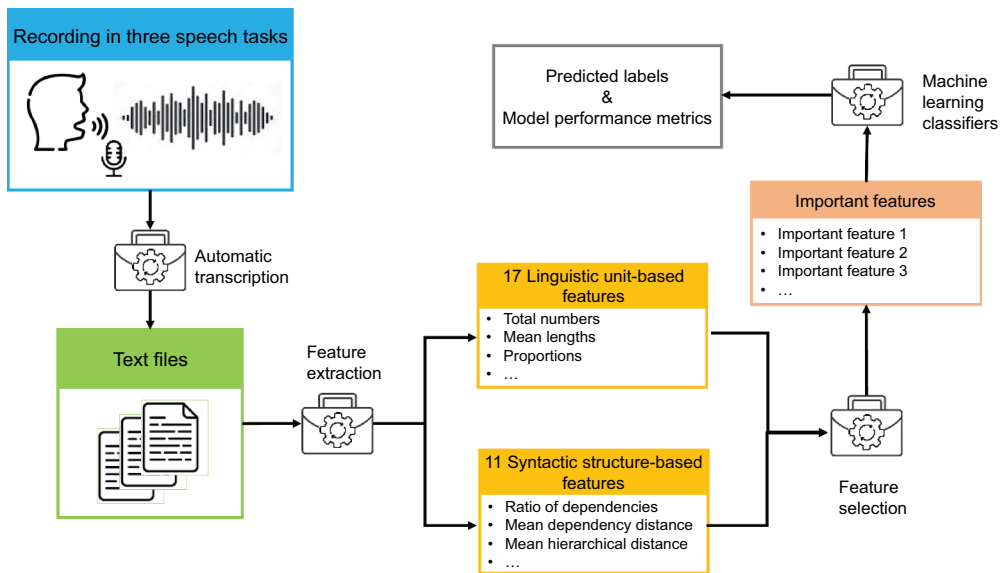


Figure 1. Overall framework of the speech data processing and analysis.

This final process is rule-based, and is therefore perfectly accurate given high-quality treebanks.

Speech tasks

To evaluate a range of discourse measures that reflect various levels of discourse representations, discourse was elicited from three types of connected speech tasks that varied in terms of stimuli and degree of organization. All tasks were administered by the same individual. Only words spoken by the participant were analyzed. Instructions for the tasks were designed with reference to the DementiaBank protocol (<https://dementia.talkbank.org/protocol/materials/>).

The picture description (PD) task was used by majority of studies to explore the linguistic impairment of individuals with cognitive decline in a relatively spontaneous, structured and well-defined context. In this study, discourse was elicited by using the “Picnic Scene” stimulus from the Western Aphasia Battery, Revised (WAB-R; Kertesz, 2006) to facilitate verbal responses and limit significant influence from working memory capacity. Participants were given a “Picnic Scene” picture and the instruction “Here is a picture. Please tell me everything you see going on in this picture”. If a participant did not give a verbal response in 10 seconds, the second prompt was given “Take a look (point to picture) and please tell me what’s happening in the picture”. If a participant gave a response of fewer than two utterances, an additional prompt was given “Anything else you can tell me about the picture?” No time-constraints were imposed.

In the story narrative (SN) task, participants were asked to make a story based on a stimulus contained six sequential pictures that were black and white line drawings. The pictures were selected from *Sanmao* comic strips created in 1940s by artist Zhang Leping (see Figure 2). Participants were given the instruction “This picture is several

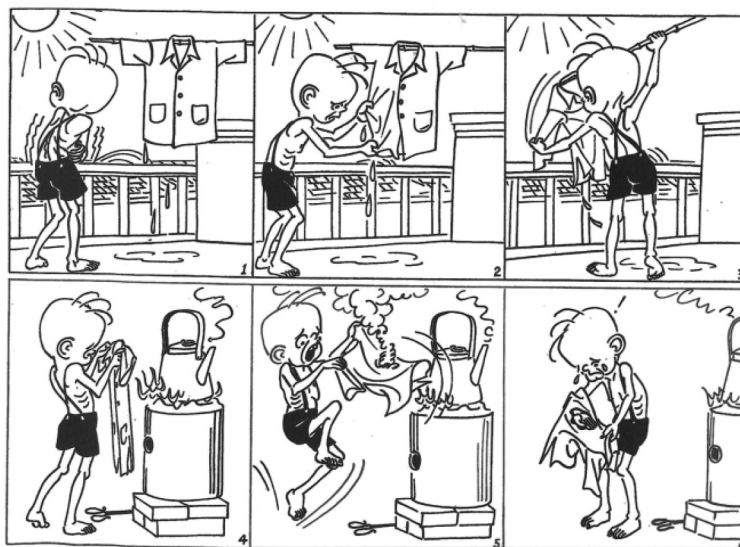


Figure 2. Sequential pictures from *sanmao* comic strips (Zhang, 2017).

pictures together in one and tell a story. Please have a look at the pictures in this order (pointing at the sequence of pictures), and then tell me the story with a beginning, a middle, and an end". Prompts similar to the picture description task would also provide when a participant did not give a verbal response in 10 seconds or gave a response of fewer than two utterances. No time limit was imposed for the task. The SN task proposed higher organizational demands on the produced story, because the organization of the story was visually provided by the sequence of pictures, eliminating the need for participants to organize their ideas before expression. This task was particularly challenging for dementia patients from a visuospatial perspective (Duong et al., 2005), largely due to the repetitive characters shifting through similar contexts while engaging in various behaviors. In addition, the SN task required a higher degree of working memory as a result of connecting each picture to previous and subsequent ones (March et al., 2006).

In the story recall (SR) task, participants were asked to retell the story of *The Cowherd and the Weaving Maid*, which is one of the most famous love stories in Chinese mythology. Participants looked through a wordless picture book within 1 minute, before describing the story. They were given the instruction "Do you remember the story of *The Cowherd and the Weaving Maid*? This picture book might remind you of how it goes. Take a look at the book, and then I'll put the book away and ask you to tell me the story in your own words". After participants looking through the book (assist with page turning if needed), and then prompted "Now please tell me as much of the story of *The Cowherd and the Weaving Maid* as you can. You can use any details you know about the story, as well as the pictures you just looked at". If a participant gave a response of fewer than three utterances, or seems to falter, allow 10 seconds, then prompted "What happened next?" or "Go on". Discourse production continued until participant concluded the story or had clearly finished.

Table 2. Corpus sizes produced by the two groups in each task.

Speech tasks	CN (N = 24)	CI (N = 25)	The entire cohort
picture description	3668	4627	8295
story narrative	3305	3406	6711
story recall	7422	8727	16149
total	14395	16760	31155

The corpus sizes are shown as sum of the text lengths in each group, measured in number of words.

Story recall was a comprehensive task requiring integrity of lexical knowledge, executive function and episodic memory in a refined and structured story frame. Given that episodic memory impairment was the most representative character of individuals with MCI, SR task had the potential to improve classification performance when discriminating MCI from typical aging. This hypothetical detection superiority was proved by some studies (Chen et al., 2024; Clarke et al., 2021; Kim et al., 2022).

Table 2 shows the corpus sizes measured in words after transcription in each speech task.

Feature extraction

In this study, we aimed to conduct a comprehensive investigation of syntactic complexity measures for detecting cognitive decline. While dozens of features were found in the literature, we excluded those which appear infrequently in our corpus and which were not applicable in terms of calculation. Table 3 provides the full list of the syntactic features extracted and the extraction methods.

All the syntactic features extracted were divided into two classes: the linguistic unit-based ones and the structure-based ones.

The linguistic unit-based linguistic features were those based on the extraction of basic linguistic units at various levels, such as sentences, T-units,⁴ clauses, phrases and words. Specifically, the features covered the frequencies, mean lengths, the proportions and ratios of the abovementioned linguistic units. A total number of 17 linguistic unit-based features were extracted. Some of these features were frequently utilized in previous research on cognitive decline detection, shown to be effective for reflecting syntactic competence and detecting cognitive decline (Fraser et al., 2016; Guinn et al., 2014; Orimaye et al., 2017).

The structure-based features concern the relations between linguistic units. In this study, we primarily focused on the dependency syntactic structure. Dependency analysis generates information about the head and dependent tokens of each dependency relation, their positions in the sentence, and the type of the dependency relation. Based on this information, we calculated the ratios of different types of dependencies and extracted four features that reflect the dependency structure, including the mean dependency distance (MDD, H. Liu et al., 2009), the mean hierarchical distance (MHD, Jing & Liu, 2015), and the dependency direction, defined as the ratio of head-initial and head-final dependencies (H. Liu, 2010). A total number of 11 syntactic structure features were extracted. Some of these, such as the MDD, were already found to be correlated with aging-related cognitive decline (J. Liu et al., 2021). Others, including MHD and dependency directions, were shown to be able to measure

Table 3. The syntactic features extracted from the transcriptions.

Domains	Sub-Domains	Features	Codes	Formulae or calculation methods
Linguistic unit-based features	Sentence	ratio of sentences	S/W	$\# \text{sentences} / \# \text{words}$
		mean length of sentences	MLS	$\sum_i \text{Len}(\text{sentence}_i) / \# \text{sentences}$
	T-units	ratio of T-units	T/W	$\# \text{T-units} / \# \text{words}$
		mean number of T-units per sentence	T/S	$\# \text{T-units} / \# \text{sentences}$
		mean length of T-units	MLT	$\sum_i \text{Len}(T - \text{unit}_i) / \# \text{T-units}$
	Clauses	ratio of clauses	C/W	$\# \text{clauses} / \# \text{words}$
		mean number of clauses per sentence	C/S	$\# \text{clauses} / \# \text{sentences}$
		mean number of clauses per T-unit	C/T	$\# \text{clauses} / \# \text{T-units}$
		mean length of clauses	MLC	$\sum_i \text{Len}(\text{clause}_i) / \# \text{clauses}$
	Phrases	proportion of noun phrases	PNP	$\sum_i \text{Len}(\text{NP}_i) / \# \text{words}$
		proportion of verb phrases	PVP	$\sum_i \text{Len}(\text{VP}_i) / \# \text{words}$
		mean number of verb phrases per T-unit	VP/T	$\sum_i \#(\text{VP in } T - \text{unit}_i) / \# \text{T-units}$
		proportion of adjective phrases	PAP	$\sum_i \text{Len}(\text{AP}_i) / \# \text{words}$
		proportion of adverb phrases	PAdvP	$\sum_i \text{Len}(\text{AdvP}_i) / \# \text{words}$
		proportion of preposition phrases	PPP	$\sum_i \text{Len}(\text{PP}_i) / \# \text{words}$
		ratio of unique words	UW/W	$\# \text{unique words} / \# \text{words}$
		ratio of repetitions	RP/W	$\text{repetitions} / \# \text{words}$
		mean depth of nodes	MDN	$\sum_i \text{depth}_i / \# \text{words}$
Syntactic structure-based features	Depth of parse trees			
	Ratio of dependencies	ratio of adverbial modifiers	advmod	$\# \text{adverbial modifiers} / \# \text{dependencies}$
		ratio of adjectival modifiers	amod	$\# \text{adjectival modifiers} / \# \text{dependencies}$
		ratio of nominal modifiers	nmod	$\# \text{nominal modifiers} / \# \text{dependencies}$
		ratio of nominal subjects	nsubj	$\# \text{nominal subjects} / \# \text{dependencies}$
		ratio of prepositional dependents	prep	$\# \text{prepositional dependents} / \# \text{dependencies}$
		ratio of compounds	cmpd	$\# \text{compounds} / \# \text{dependencies}$
	Structure of dependencies	mean dependency distance	MDD	$\sum DD / \# \text{dependencies}$
		mean hierarchical distance	MHD	$\sum HD / \# \text{dependencies}$
		ratio of head final dependencies	HF/W	$\# \text{head final dependencies} / \# \text{words}$

$\square$: the number of $\square$.
Len $\square$: length of $\square$, i.e., the number of words in $\square$.
 $\square_i$: the i th $\square$ in the transcription.
Unique words refer to different word types.
Repetitions are identified for each sentence. The number of repetitions in one sentence is the number of repeated words, calculated as the difference between the total number of words in that sentence and the number of unique words in that sentence.
P $\square$: the proportion of a linguistic unit, defined as the sum of length of all of its occurrences in the corpus divided by the length of the corpus (total number of words).
DD: the (horizontal) dependency distance of a dependency relation.
HD: the (vertical) hierarchical distance of a dependency relation.
cmpd: compound

syntactic complexity in second language acquisition, writing assessment, as well as translation and interpreting (Fan & Jiang, 2019; Xu & Liu, 2023). Therefore, we also intended to apply this measure to examine whether it can reflect cognitive decline in older adults.

Statistical analysis and machine learning algorithms

To identify features with high discriminative power as potential biomarkers for cognitive impairment, a series of statistical analyses were conducted. First, we performed two-sided Mann-Whitney U tests to compare the feature distributions between the cognitively impaired and cognitively normal participants. In addition, we calculated the AUC values for each feature to quantify their ability to distinguish between the two groups. The results of these analyses, including p-values and AUC values, were then visualized using a volcano plot. This approach follows standard practices in biomarker studies (e.g., Y. Guo et al., 2024), where features are assessed based on both statistical significance and effect size. We set the significance level at 0.05 to determine which features showed meaningful group differences.

Then machine learning methods along with feature selection algorithms were employed to search for composite feature, or feature panel (i.e., feature combinations). Before that, we performed z-score normalization (sample values of each feature were subtracted from their mean value and then divided by their standard deviation) to increase the stability of training process and enhance the model performance. Three linear machine learning classifiers, including Logistic Regression (LR), Support Vector Machines (SVM), and Linear Discriminant Analysis (LDA), were applied due to their interpretability in capturing the relationships between the final output composite feature and the basic input features.

For feature selection, we employed two schemes. The first scheme involved using the Recursive Feature Elimination (RFE) algorithm on the entire feature set to evaluate model performance with varying numbers of features and select the optimal subset, resulting in the so-called “large panel”. RFE is an efficient algorithm which eliminate the least important feature each step based on the ranking of features, but its limitation lies in its path dependent nature, that is, it identifies only locally optimal solutions in the feature elimination sequence, rather than the global optimum. The second scheme applied an exhaustive search using the brute-force Exhaustive Feature Selection (EFS) algorithm to find the best feature combinations within the scope of the highly discriminative single-dimension features identified for each task. The resultative feature subsets produced following this scheme was called the “small panel”. The performances of the models with different speech tasks, classifiers and feature selection schemes were compared. Our ultimate goal was to identify the optimal feature combination that maximizes performance while using the minimal number of base features.

We were aware that multicollinearity among features could potentially affect model performance. However, since the feature selection algorithm itself is designed to select optimal feature combinations based on performance, we did not exclude any features a priori. Instead, we relied on the feature selection process to eliminate those with multicollinearity. The pairwise correlation heatmap for the 27 features across the three tasks is presented in [Figure A1](#).

To prevent overfitting, we adopted the Stratified Leave-Pair-Out Cross-Validation (SLPOCV) for training and validation, following Roark et al. (2011), Orimaye et al. (2017), and Fraser et al. (2019). In each loop, the selected feature values of a pair of participants (including one positive and one negative) were taken as the test set, while the rest as the training set. The trained model was then used to predict the label of the held-out sample.

The model performance was primarily measured by the mean accuracy and AUC in cross validation, while we also provided values for sensitivity, specificity, and F1-score. Since the dataset is basically balanced, we simply reported accuracy (ACC), but not F1-score.

All visualizations were generated using packages in R (version 4.4.1) and Python (version 3.12.6), depending on which was more suitable.

Results

Discriminative power of single-dimensional syntactic features

Figure 3 presents the p -values obtained from two-sided Mann-Whitney tests, and the discriminability of each measure between cognitively healthy and cognitively impaired older adults indicated by AUC values. For the full list of descriptive statistics, p -values and AUC values of each syntactic complexity measure with significant p -values and AUC values bolded, see Table A2.

It can be observed that most features are positioned to the left of the vertical dashed line, indicating a negative correlation with cognitive impairment, consistent with the understanding of reduced syntactic complexity in individuals with cognitive impairment. Features on the right may, in fact, suggest a form of “incomplexity”. Among the highly discriminative features, only the proportion of adverbial modifiers is located on the right. In what follows, we present $AUC_{adjusted} (= 1 - AUC_{original})$ when the original AUC is less than 0.5, ensuring that all AUC values are consistently between 0.5 and 1 and reflects differentiability between groups.

In addition, it is shown that no common feature manifests high discriminability across all three tasks. The proportion of adjective phrases (PD: $p = 0.012$, $AUC = 0.708$; SR: $p = 0.019$, $AUC = 0.695$) and the ratios of head final dependencies (SN: $p = 0.011$, $AUC = 0.713$; SR: $p = 0.042$, $AUC = 0.670$) have high discriminability in two tasks, but not significant in the third. In addition, there are also task-specific important single features. For the picture description task, the MCI and HC groups differed significantly on mean dependency distance ($p = 0.014$, $AUC = 0.705$), ratio of compounds ($p = 0.016$, $AUC = 0.702$), proportion

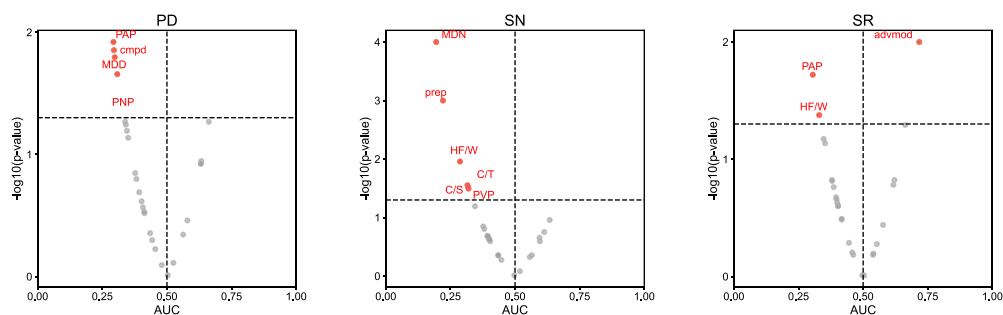


Figure 3. Volcano plots showing the $-\log_{10}(p\text{-value})$ (y axis) and AUC (x axis) for the single-dimensional syntactic features in the three tasks. Features above the horizontal dashed black line were significantly associated with cognitive impairment in each task, denoted in red dots. Features to the left of the vertical dashed line are negatively correlated with cognitive impairment (CI), indicating that their values are lower in the CI group compared to the healthy control (CN) group, while those on the right show the opposite pattern.

of noun phrases ($p = 0.022$, $AUC = 0.692$). For the story narrative task, the CI group had a lower mean number of clauses per sentence ($p = 0.030$, $AUC = 0.682$), mean number of clauses per T-unit ($p = 0.028$, $AUC = 0.684$), proportion of verb phrases ($p = 0.032$, $AUC = 0.680$), mean depth of nodes ($p = 0.000$, $AUC = 0.805$), ratio of prepositional dependents ($p = 0.001$, $AUC = 0.779$) than the CN group. As for the story recall task, U tests showed that the CI and CN groups differed significantly on proportion of adjective phrases ($p = 0.019$, $AUC = 0.695$), and ratios of adverbial modifier dependencies ($p = 0.010$, $AUC = 0.717$). Among all important feature with high discriminability between cognitively impaired and cognitively healthy older adults, mean depth of nodes has the highest AUC of 0.805, followed by the ratio of compounds with $AUC = 0.779$, both in story narration task. Overall, it is indicated that the discriminability of syntactic features interacts with the speech task type, and there is no universal feature shared by all the tasks.

Machine learning classification

In the machine learning-based search for discriminative composite features for cognitive impairment, the current study compared two feature selection schemes. One involved performing the brute-force method of Exhaustive Feature Selection (EFS) within the scope of highly discriminative single-dimensional features ("small panel"), while the other applied the Recursive Feature Elimination (RFE) method to the entire feature set to find the optimal number of features n as a hyperparameter ("large panel").

Figures 4 and 5 show how the average accuracy and AUC changes with the number of features respectively in the case of small panels and large panels. Both figures illustrate that using all highly discriminative features often performs worse than using only a subset of them, suggesting the necessity of applying feature selection methods. Figure 4 demonstrates that, within a narrow search range, the optimal number of features for the three tasks is 2 (PD: cmpd, MDD), 2 (SN: MDN, prep), and 3 (SR: PAP, advmod, HF/W), respectively. In addition, as seen in Figure 5, many models and feature combinations exhibit distinct performances in terms of the two evaluation metrics, accuracy and AUC. A high accuracy ensures the model's generalization ability and prevents overfitting, while AUC reflects the differentiability of the (composite) feature between groups. If The model with the highest AUC was achieved by an LDA model with 12 features in the picture description task, reaching the cross-validation AUC value of 0.890, but its mean accuracy is only 0.765. When both metrics are considered, the thresholds were set at $ACC = 0.8$ and $AUC = 0.85$. For the picture description task, only a 6-feature panel with the SVM classifier met both conditions, whereas an 11-feature panel with SVM and a 14-feature panel with LDA approach the criteria. For the story recall task, none of the feature combinations meet the threshold, while a 13-feature panel with SVM is the closest. For the story narrative task, several feature combinations meet the threshold, within which a 2-feature panel with the LR model showing the best performance while also using the fewest features. The details of the selected features are provided in Table A3. Note that several relatively highly discriminant uni-dimensional features for cognitive decline detection might not be selected in the multi-dimensional models. Table A4 displays the full results of model performance, measured by the validation accuracy, precision, and sensitivity as well as area under curve (AUC). Overall, the models generally performed best on the story narrative task, followed by the picture description task, with the story recall task showing the lowest performance.

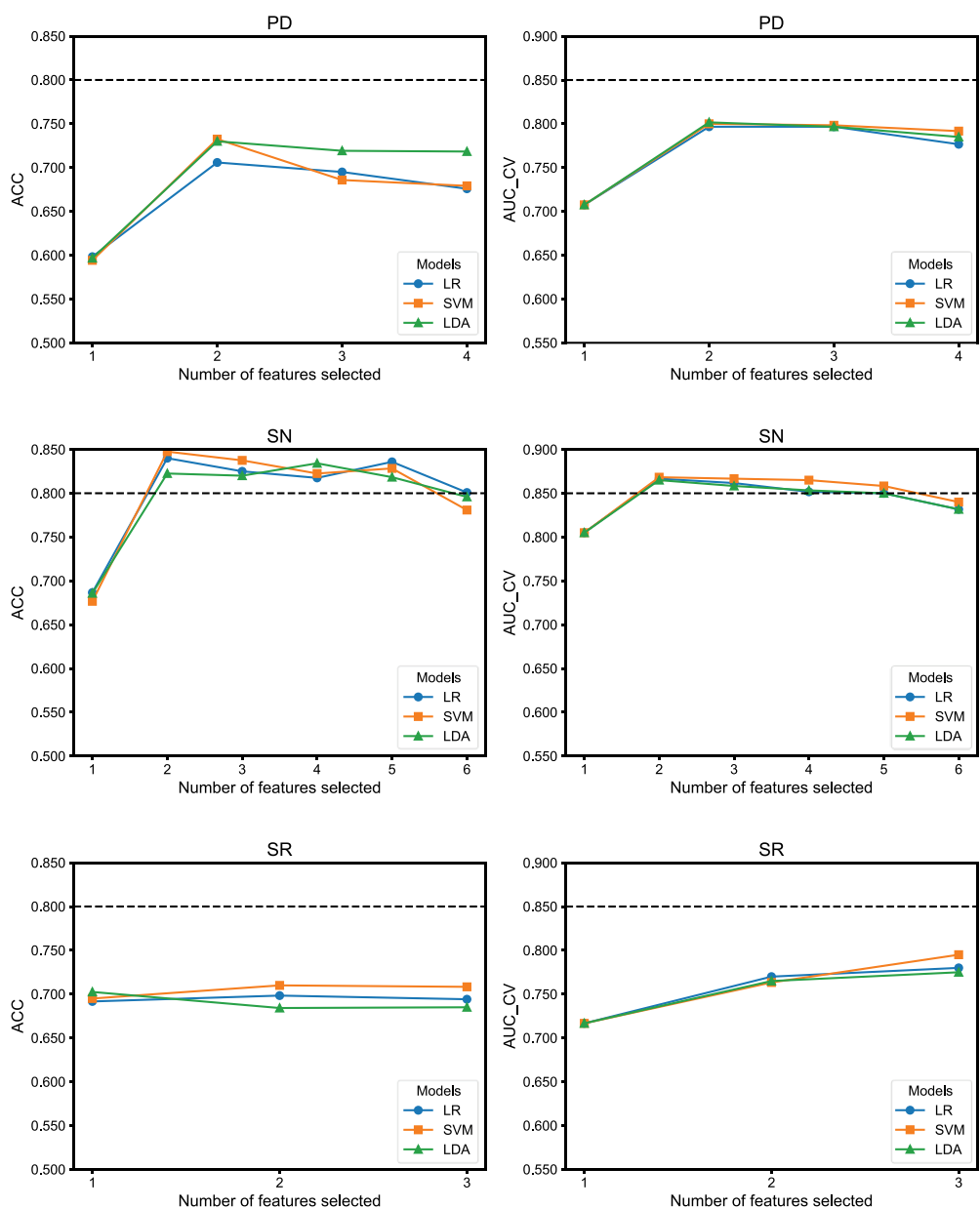


Figure 4. The average accuracy and AUC given different numbers of features to be selected by the EFS method within the scope of highly discriminative single-dimensional features. First row: the picture description task. Second row: the story narrative task. Third row: the story recall task. Left column: the mean accuracy from SLPO cross-validation. Right column: the mean AUC from SLPO cross-validation.

Figure 6 further shows the ROC curves of highly discriminative single-dimensional features, small and large feature panels with the highest AUC in the three tasks. It is indicated that the feature combinations learned by the machine learning models significantly improved the discriminability between the CN and CI groups, as measured by AUC. It has been verified that no multicollinearity exists among the selected feature panels.

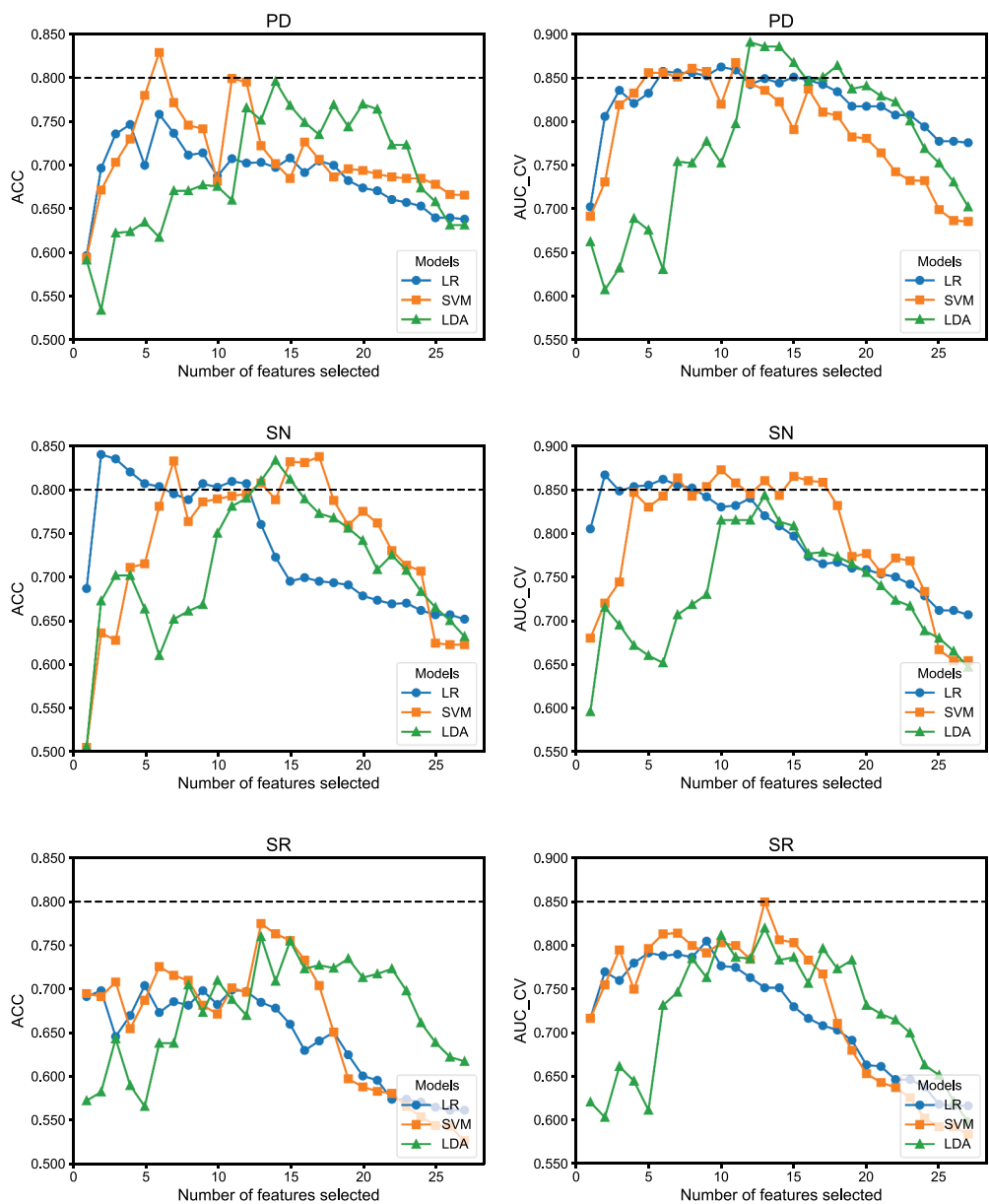


Figure 5. The average accuracy and AUC given different numbers of features to be selected by the RFE method in the full feature set. First row: the picture description task. Second row: the story narrative task. Third row: the story recall task. Left column: the mean accuracy from SLPO cross-validation. Right column: the mean AUC from SLPO cross-validation.

Discussion

The objective of the present study was to search for highly discriminative syntactic complexity biomarkers of cognitive impairment and early stages of dementia. A Mandarin corpus consisting of speech transcriptions from native elderly speakers

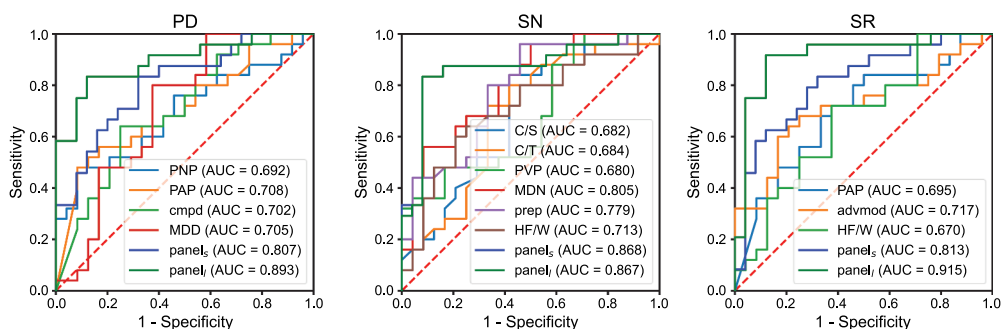


Figure 6. The ROC curves of highly discriminative single dimensional features, small and large feature panels in three tasks.

based on three speech tasks was employed to detect cognitive decline. Specifically, we explored the potential of linguistic unit-based and syntactic structure-based features in classification for distinguishing between CI and CN, both in terms of individual features and composite features generated by machine learning models. In the following, we discuss the research questions in turn.

The first research question addresses which models and feature subsets demonstrate high discriminability between CI and CN populations. The current study has obtained several models with certain feature combinations that reach a mean accuracy higher than 0.80 and a cross-validation AUC higher than 0.85. These AUC values of the composite features are all higher than any single syntactic feature that we have found in any task, suggesting that the employment of machine learning methods has significantly enhanced the detection of cognitive impairment. Our best results are basically within the first tier of the state-of-the-art models for cognitive decline detection with Mandarin speech, with accuracy and AUC around 0.83 (Chien et al., 2019, AUC = 0.838; N. Liu et al., 2022, ACC = 83.7%; Ying et al., 2023, ACC = 89.1% for long audios and 84.0% for short audios; R. Wang et al., 2023, ACC = 81%, AUC = 0.81). Indeed, our results were still lower than T. Wang et al. (2021), who also employed three speech tasks and obtained a rather excellent result. Their best performances reached an accuracy of 0.96 with soft voting as task fusion method. As for each specific task, the best performance was achieved on their picture description task, still approaching 90%. A possible reason could be that they used a more diverse set of features, not limited to syntactic ones, or that they identified more important and intrinsic features regarding specific tasks than us. Despite this, our results remain sufficiently satisfactory.

The second research question discusses which syntactic feature(s) play significant roles among these models. The study revealed several single-dimensional features and feature combinations with high discriminability in each task. All these selected discriminative measures fall into two types. The first type at the macro-level, such as mean dependency distance (MDD), ratio of head final dependencies (HF/W) and mean depths of nodes (MDN), are more robust in terms of measurement, since they are calculable in all cases, regardless of the length of the transcribed texts. On the contrary, those features focusing on specific linguistic units (e.g., the proportions of adjective phrases) might not be comparable due to the flooring effect in short texts as their values in both groups approach zero. In other words, certain classes of construction might not appear in certain short speech recordings if they

were not designed to be elicited. Among the macro-level measures, the mean depth of nodes with the highest AUC in the story narrative task, to our knowledge, is a newly identified marker and has not been reported in previous literature, which merits further verification. This measure reflects the speakers' ability to compose complex sentences with deep syntactic trees, i.e., using more phrasal nodes and embedding structures. As for the ratio of head final dependencies, also known as dependency direction (H. Liu, 2010) or so-called Liu-directionality (Fisch et al., 2019), it was found to be discriminative in two tasks (SN and SR). J. Liu et al. (2024) compared the values of a similar measure for healthy English and Mandarin-speaking older adults, but they did not investigate its change due to cognitive decline for the Mandarin-speaking population in that study or its preceding similar study (J. Liu et al., 2021). Finally, mean dependency distance (MDD) in the CI group was significantly lower than the CN group in the picture description task, but not in story narrative and story recall. This is somewhat in line with previous findings in that whether MDD is a significant marker of cognitive impairment depends on speech elicitation tasks. MDD is a common operationalization of syntactic complexity applied in multiple fields, and has been shown to a key textual measure reflecting cognitive abilities relating to memory (H. Liu, 2008). It has been found to be a significant predictor of cognitive impairment in delayed story retelling (Roark et al., 2007, 2011), written narratives (Sand Aronsson et al., 2021), casual conversation (J. Liu et al., 2021), and several semi-spontaneous tasks (Calzà et al., 2021), while there is no statistical difference between the groups in immediate story telling (Roark et al., 2007, 2011) and picture description narrative (Lundholm Fors et al., 2018). Specifically, Lundholm Fors et al. (2018), Calzà et al. (2021), and Sand Aronsson et al. (2021) all made use of the specific task of Cookie theft picture description, but reached contradictory results. Possible explanations include the calculation method of the measure, the language, and the mode (written or spoken), as these variables vary in the abovementioned studies. In sum, future research calls for more standardized measurement and more rigorous comparative studies focusing on the effect of one factor at a time while controlling for other variables.

The third research question aims to determine whether there are any features or feature combinations that exhibit high discriminability across different tasks and classifiers. On the one hand, no features change significantly due to cognitive decline across all three tasks. Rather, each speech task prefers different features or feature combinations, indicating that measures of syntactic complexity used in this study might not be universal but task-specific. This is in line with a number of previous findings (Clarke et al., 2021; Roark et al., 2011; T. Wang et al., 2021). In other words, it seems that there is coupling between specific features and speech tasks. For instance, T. Wang et al. (2021) selected different feature sets for three speech tasks and all achieved decent results, suggesting that their approach captured critical features for each task. Possible causes are associated with the intrinsic characteristics of the speech task. The picture description and story narrative tasks provided explicit instructions and consistent materials, allowing participants to generate their responses from similar mental representations, reducing variation in narrative content. In contrast, the story recall task required participants to independently reconstruct the story from memory, leading to greater variability in linguistic features due to differences in recall ability and interpretation. This variability, especially in spontaneous, independent narration, introduces challenges for ML models in isolating features related specifically to cognitive impairment.

On the other hand, several small feature panels demonstrated optimal performance across different classifiers in each corresponding task, whereas for large feature panel,

none well-performing feature combinations were consistently selected by different classifiers. Since the three classifiers we employ are all simple linear models, aimed at finding linear combinations of basic features, feature sets that demonstrate high discriminative power across classifiers may reveal key linguistic biomarkers of cognitive impairment.

The last research question discusses how to determine the optimal feature combination(s) which might be used for cognitive screening and clinical diagnosis. Taking everything into account, the best model seems to be the two-feature panel (MDN, prep) used in the story narrative task with a relatively balanced performance between accuracy and AUC ($ACC = 0.840$, $AUC = 0.867$). Since we used features with specific linguistic significance rather than black-box embedding features, and simple linear machine learning models rather than complex neural network models, our models outperform most of the previous models (Chien et al., 2019; N. Liu et al., 2022; Tsai et al., 2021; Yang et al., 2022) in term of both interpretability and simplicity. Moreover, given that the 2-feature panel performs best regardless of classifiers, we might have touched upon key linguistic biomarkers underlying the mechanisms of cognitive decline. To sum up, this two-feature combination in the story narrative task was considered the most effective and cost-efficient, making it the most suitable candidate for inclusion in future cognitive screening battery.

This study no doubt has several limitations. First, the relatively small sample size used in this research poses challenges for the validity and the generalizability of feature selection. A larger, more substantial, and more diverse dataset is needed to obtain a more stable model with stronger generalization performance. In addition, future studies might adopt two or more independent datasets for cross-validating the effectiveness of the features selected. Second, the quality of transcriptions within the corpus is another concern. Visual inspection revealed that many segments marked by full stops do not form complete sentences, with participants often providing filler words or fragmented responses, especially in the picture description task. This may reflect broader issues inherent in spoken corpora. In addition, the variation in speaking styles observed in the recordings – where both CN and CI participants sometimes produced isolated words rather than coherent sentences – could affect the linguistic features extracted and, in turn, influence the ML model's performance. Thirdly, RFE is a path-dependent feature selection method, which means it does not exhaust and compare the performances of all the combinations of features given the number of features to be selected. To overcome this deficiency, better feature selection method might be adopted in the future research. Finally, researchers might generalize the findings to other languages similar to Mandarin, such as Cantonese, Hokkien. Expanding the scope to these languages would allow for a broader understanding of how syntactic complexity measures and machine learning models perform across different linguistic structures, further enhancing the cross-linguistic applicability of the findings.

Conclusions

In conclusion, this study investigated the potential of using machine learning method to search for highly discriminative linguistic biomarker of cognitive decline on a spoken corpus of Mandarin-speaking older adults with three speech tasks. We focused on syntactic features extracted from the transcriptions of the participants' speech production and identified features that are critical for classification, such as mean dependency distances and mean depth of nodes. We showed that a 2-feature panel (mean depth of nodes, and the proportions

of prepositional dependencies) across three linear machine learning models in the story narrative task achieved the most remarkable classification performance. Our study confirms the potential of textual features as low-cost, non-invasive linguistic biomarkers for cognitive impairment, particularly in community-based screening settings. Future research should expand the sample size and include more external cohorts to validate the identified markers.

Notes

1. AUC is a common metric to quantify the discriminant power of a classifier between two groups, ranging between 0.5 and 1. Value 1 represents fully discriminable and value 0.5 represents the chance level for differentiating between two groups. It was previously used mostly in the machine learning approach. Yet in fact it can also be used to quantify the degree of (non-)overlapping between two distributions on any certain single feature dimension, by taking different cutoff values to form a classifier. Currently it has become the main metric to report in the biological and medical literature in search of biomarkers.
2. <https://ageing.tongji.edu.cn/info/1006/1233.htm>
3. All Python codes for text preprocessing, feature computation, and machine learning have been uploaded to GitHub and are publicly available (<https://github.com/Yihtsy/Dementia-Syntax-ML/>).
4. Among these, the T-unit is a somewhat less mentioned concept in traditional grammar yet widely applied now in applied linguistics. A T-unit refers to an intermediate level of unit between sentence and clauses, where coordinated clauses are counted as different T-units, while subordinated clauses with their corresponding main clauses are taken as a whole and regarded as single T-units (Hunt, 1970).

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The original speech data used in this study cannot be publicly shared due to project confidentiality restrictions. However, the extracted feature data used in the analyses can be made available upon reasonable request by contacting the first author.

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Appendices

Table A1. Criterion for deciding the level of cognitive ability of a subject based on MoCA-B scores.

Cognitive ability level	Highest education level (Years of education)		
	Primary school (≤6)	Middle school (7–12)	University or college (≥13)
CN	20–30	23–30	25–30
MCI	14–19	16–22	17–24
Mild AD	11–13	12–15	14–16
Moderate or severe AD	≤10	≤11	≤13

Table A2. The statistics of syntactic features for each group in three speech tasks.

Features	PD				SN				SR			
	Median (IQR)		<i>p</i>	AUC	Median (IQR)		<i>p</i>	AUC	Median (IQR)		<i>p</i>	AUC
	CN	CI			CN	CI			CN	CI		
S/W	0.05 (0.05)	0.07 (0.04)	.054	.662	0.05 (0.04)	0.05 (0.03)	.254	.596	0.04 (0.04)	0.05 (0.04)	.150	.621
C/W	0.20 (0.03)	0.20 (0.04)	.772	.525	0.21 (0.03)	0.20 (0.05)	.441	.565	0.18 (0.02)	0.19 (0.03)	.529	.553
T/W	0.09 (0.03)	0.09 (0.03)	.453	.563	0.06 (0.03)	0.07 (0.02)	.177	.613	0.06 (0.02)	0.07 (0.03)	.165	.617
UW/W	0.51 (0.13)	0.45 (0.15)	.242	.598	0.53 (0.16)	0.54 (0.11)	.826	.519	0.44 (0.10)	0.43 (0.14)	.976	.503
RP/W	0.24 (0.10)	0.21 (0.10)	.204	.607	0.27 (0.21)	0.25 (0.12)	.441	.565	0.26 (0.11)	0.20 (0.16)	.516	.555
C/S	3.86 (3.98)	2.78 (1.25)	.057	.659	5.00 (4.19)	3.83 (2.00)	.030*	.682	4.21 (3.23)	3.38 (3.19)	.250	.597
T/S	2.00 (2.13)	1.67 (0.87)	.064	.655	2.06 (1.63)	1.75 (1.25)	.207	.606	2.07 (1.86)	1.91 (0.95)	.327	.583
VP/T	8.44 (8.08)	6.21 (3.64)	.073	.650	13.27 (13.59)	11.79 (8.32)	.211	.605	11.17 (9.81)	7.93 (11.09)	.624	.542
C/T	2.27 (1.00)	2.28 (0.54)	.503	.557	3.30 (1.56)	2.67 (0.48)	.028*	.684	2.89 (0.91)	2.61 (1.05)	.219	.603
MLS	18.68 (19.63)	13.90 (8.10)	.054	.662	20.25 (23.49)	20.00 (10.30)	.254	.596	23.87 (16.92)	18.45 (14.62)	.150	.621
MLC	9.18 (2.87)	8.68 (2.25)	.303	.587	10.27 (2.06)	9.69 (2.23)	.156	.619	10.58 (3.62)	9.69 (2.76)	.067	.653
MLT	11.47 (4.35)	10.83 (2.46)	.271	.592	15.22 (5.27)	13.46 (3.70)	.142	.623	15.66 (5.29)	13.47 (5.45)	.211	.605
PNP	0.60 (0.14)	0.52 (0.18)	.022*	.692	0.46 (0.16)	0.41 (0.17)	.064	.655	0.59 (0.19)	0.54 (0.10)	.234	.600
PVP	1.01 (0.68)	0.88 (0.47)	.142	.623	1.48 (0.50)	1.43 (0.60)	.032*	.680	1.25 (0.34)	1.19 (0.44)	.653	.538
PAP	0.02 (0.02)	0.01 (0.02)	.012*	.708	0.00 (0.01)	0.01 (0.01)	.472	.557	0.01 (0.01)	0.00 (0.01)	.019*	.695
PAdvP	0.06 (0.05)	0.06 (0.04)	.596	.545	0.08 (0.04)	0.08 (0.06)	.234	.600	0.08 (0.02)	0.10 (0.04)	.051	.663
PPP	0.07 (0.08)	0.11 (0.16)	.119	.631	0.08 (0.06)	0.07 (0.06)	.204	.607	0.10 (0.05)	0.10 (0.10)	.976	.503
MDN	4.94 (1.21)	4.73 (0.92)	.294	.588	5.68 (0.84)	5.17 (0.53)	.000***	.805	5.69 (0.79)	5.26 (0.94)	.153	.620
advmod	0.02 (0.03)	0.03 (0.02)	.113	.632	0.04 (0.02)	0.03 (0.05)	.529	.553	0.04 (0.02)	0.06 (0.03)	.010*	.717
amod	0.00 (0.01)	0.00 (0.01)	.805	.520	0.00 (0.00)	0.00 (0.00)	.968	.503	0.01 (0.01)	0.00 (0.01)	.073	.647
nmod	0.04 (0.02)	0.04 (0.03)	.976	.503	0.02 (0.02)	0.02 (0.02)	.439	.565	0.04 (0.02)	0.05 (0.02)	.653	.538
nsubj	0.09 (0.03)	0.12 (0.03)	.119	.631	0.09 (0.03)	0.10 (0.04)	.110	.634	0.09 (0.03)	0.10 (0.03)	.363	.577

(Continued)

Table A2. (Continued).

Features	PD				SN				SR			
	Median (IQR)		<i>p</i>	AUC	Median (IQR)		<i>p</i>	AUC	Median (IQR)		<i>p</i>	AUC
	CN	CI			CN	CI			CN	CI		
prep	0.04 (0.03)	0.04 (0.02)	.347	.579	0.06 (0.04)	0.03 (0.03)	.001**	.779	0.04 (0.02)	0.05 (0.02)	.638	.540
cmpd	0.02 (0.02)	0.02 (0.02)	.016*	.702	0.00 (0.01)	0.00 (0.01)	.221	.595	0.01 (0.01)	0.01 (0.01)	.173	.614
MDD	2.84 (0.99)	2.54 (0.38)	.014*	.705	3.29 (1.31)	3.16 (0.76)	.234	.600	3.30 (0.99)	3.17 (1.24)	.322	.583
MHD	2.27 (0.63)	2.13 (0.47)	.441	.565	2.40 (0.52)	2.35 (0.57)	.234	.600	2.69 (0.76)	2.36 (0.75)	.250	.597
HF/W	0.60 (0.06)	0.56 (0.07)	.159	.618	0.59 (0.07)	0.54 (0.07)	.011*	.713	0.60 (0.04)	0.59 (0.05)	.042*	.670

IQR: interquartile range. AUC: area under the Receiver Operating Characteristic curve. Two-sided Mann-Whitney U tests were performed with significance levels for *p*-values (*: *p* < 0.05, **: *p* < 0.01, ***: *p* < 0.001). The *p*-values less than 0.05 and the corresponding AUC values were bolded.

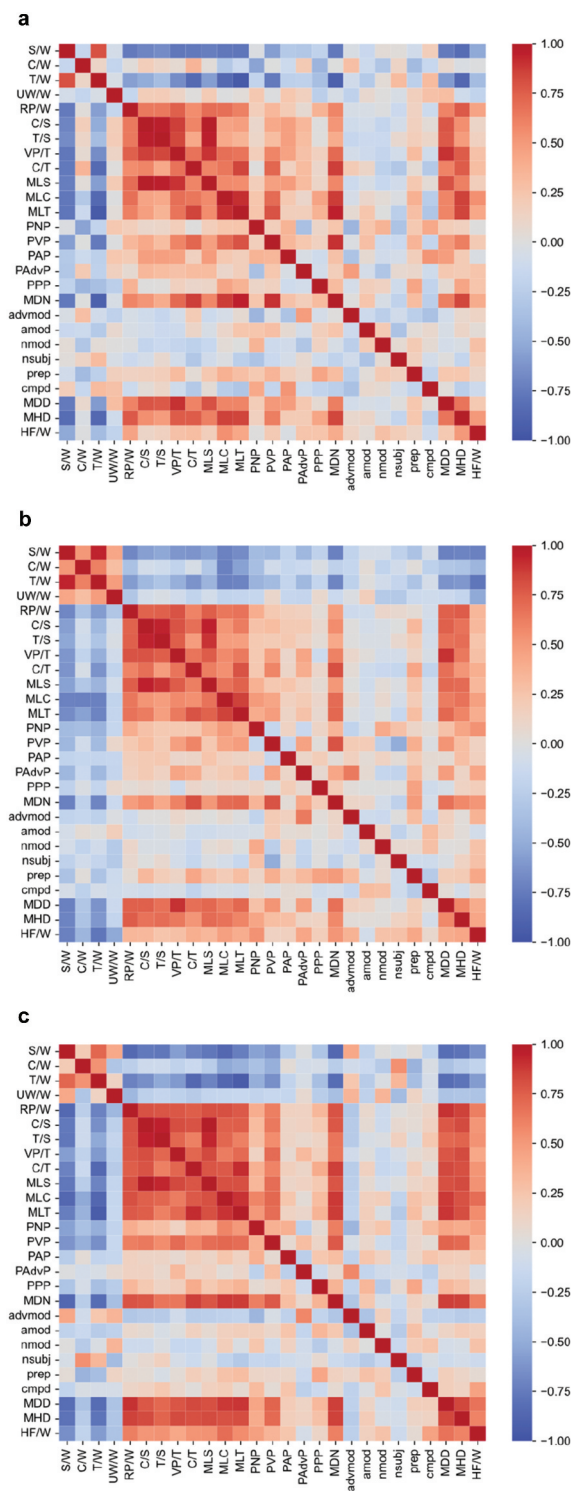


Figure A1. Heatmaps of pairwise correlations between all the extracted features for the three speech tasks. (A) picture description (PD); (B) story narrative (SN); (C) story recall (SR).

Table A3. Best feature panels selected in the three speech tasks.

Speech tasks	Small panel	Large panel
PD	across all three classifiers, 2 features: cmpd, MDD	SVM, 6 features: PNP, PAdvP, PPP, advmod, nsubj, MDD
SN	across all three classifiers, 2 features: MDN, prep	SVM, 10 features: T/W, T/S, VP/T, PNP, PVP, PAdvP, advmod, cmpd, MDD, HF/W
SR	across all three classifiers, 3 features: PAP, advmod, HF/W	SVM, 13 features: RP/W, MLS, MLC, PVP, PAP, PPP, MDN, advmod, nmod, nsubj, prep, MDD, HF/W

Table A4. Classification performances of highlighted models in each speech task.

Task	Feature set	Model	# Features	ACC	REC	SPEC	F1	AUC _{CV}	AUC _{self}
PD	HD	LR	4	0.676	0.742	0.523	0.649	0.777	0.813
		SVM	4	0.679	0.722	0.539	0.628	0.792	0.818
		LDA	4	0.713	0.813	0.560	0.708	0.785	0.820
	SP	LR	2	0.706	0.785	0.553	0.683	0.797	0.807
		SVM	2	0.733	0.840	0.575	0.735	0.800	0.812
		LDA	2	0.730	0.835	0.573	0.730	0.802	0.807
	LP	LR	10	0.687	0.702	0.566	0.621	0.862	0.932
		SVM	6	0.828	0.832	0.757	0.784	0.855	0.893
		LDA	12	0.765	0.817	0.648	0.738	0.890	0.965
SN	HD	LR	6	0.801	0.803	0.718	0.748	0.832	0.862
		SVM	6	0.781	0.788	0.684	0.724	0.840	0.862
		LDA	6	0.796	0.765	0.729	0.721	0.832	0.868
	SP	LR	2	0.840	0.845	0.768	0.797	0.867	0.867
		SVM	2	0.848	0.857	0.776	0.809	0.868	0.868
		LDA	2	<u>0.823</u>	0.798	0.757	0.754	<u>0.865</u>	0.870
	LP	LR	2	<u>0.840</u>	0.845	0.768	0.797	<u>0.867</u>	0.867
		SVM	10	0.789	0.800	0.685	0.737	0.873	0.943
		LDA	13	0.810	0.783	0.743	0.744	0.843	0.952
SR	HD	LR	3	0.694	0.710	0.576	0.631	0.780	0.812
		SVM	3	0.708	0.700	0.603	0.630	0.795	0.813
		LDA	3	0.685	0.680	0.577	0.608	0.775	0.810
	SP	LR	3	0.694	0.710	0.576	0.631	0.780	0.812
		SVM	3	0.708	0.700	0.603	0.630	0.795	0.813
		LDA	3	0.685	0.680	0.577	0.608	0.775	0.810
	LP	LR	9	0.698	0.760	0.557	0.666	0.805	0.903
		SVM	13	0.775	0.762	0.695	0.708	0.850	0.915
		LDA	13	0.760	0.758	0.677	0.703	0.820	0.953

HD: the subset of all highly discriminative single-dimensional features. SP: small feature panel selected from HD. LP: large feature panel selected from the entire feature set. ACC: accuracy. REC: recall or sensitivity. SPEC: specificity. F1: the F score. AUC: area under the ROC curve. AUC_{CV}: the mean AUC across stratified leave-pair-out cross-validations. AUC_{self}: the AUC value of the distribution of two groups along the dimension of the final composite feature. Bold: the best model for each task. Underlined: AUC_{CV} > 0.85 and ACC > 0.80.